DATA ENGINEERING

Snowflake & BigQuery Visual Guide

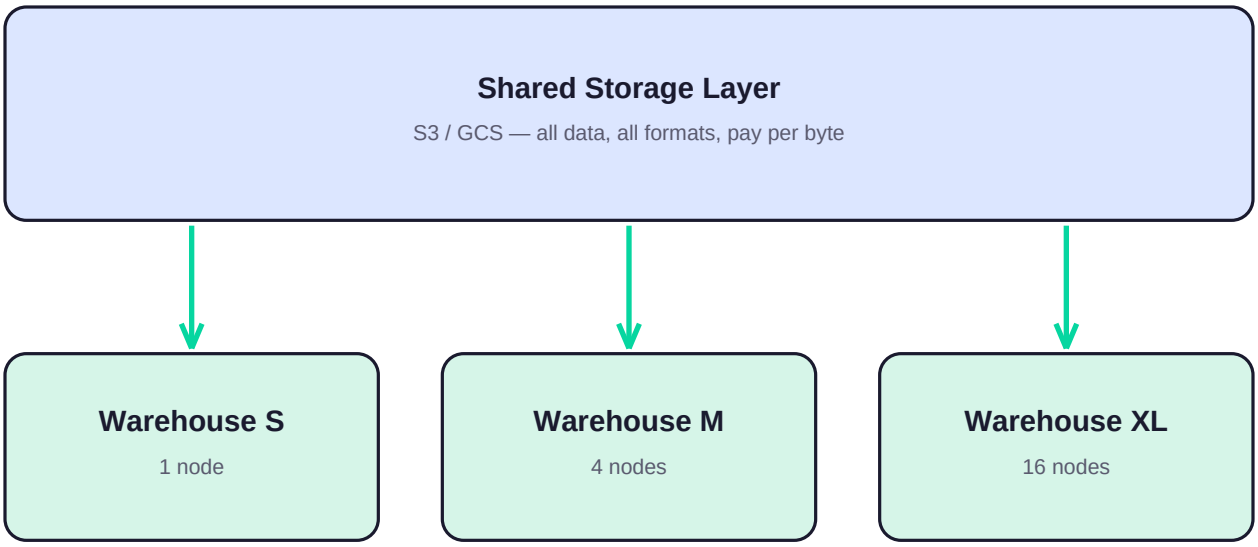
Decoupled storage and compute.
How cloud warehouses scale elastically.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Decoupled Storage & Compute

The core idea: store once cheaply, spin compute up and down independently.

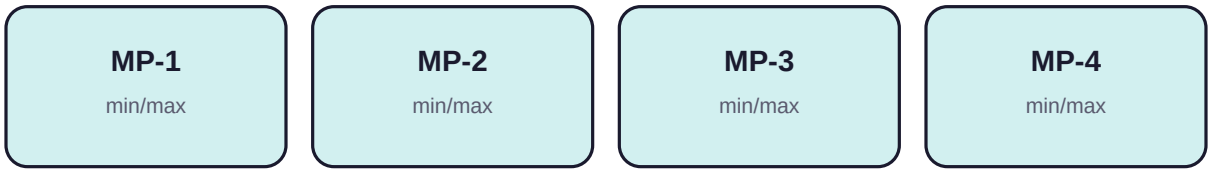


Compute clusters (virtual warehouses) attach to the same storage, run the query, then suspend. No data to move, no cluster to keep warm.

STEP 2/6

Micro-Partitions: Automatic Layout

Snowflake slices tables into ~16MB immutable, columnar micro-partitions with metadata.



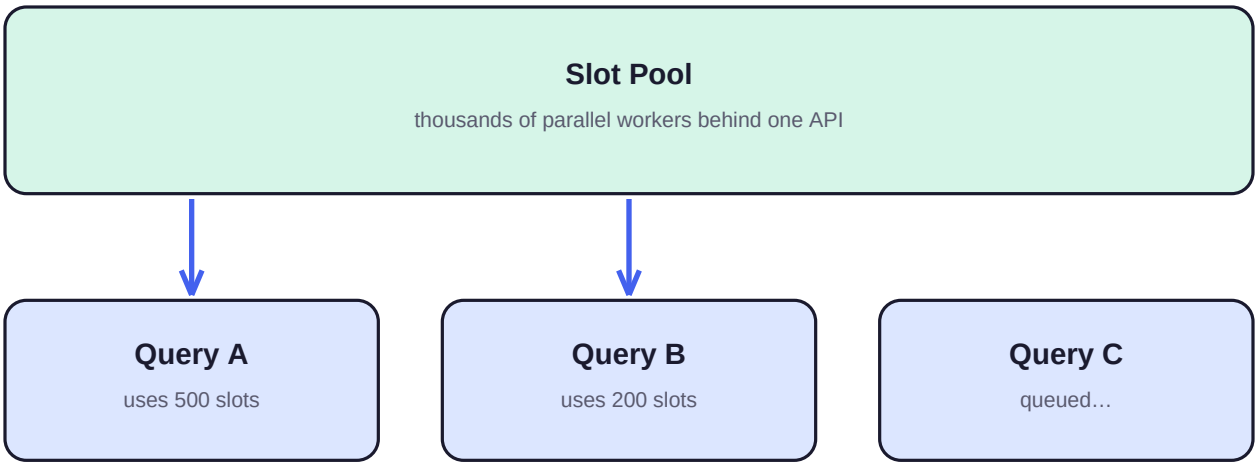
Each micro-partition stores column min/max, null counts, distinct counts.
Query: WHERE date = '2026-10-01' → the optimizer skips
partitions whose range can't match. No manual partitioning needed.

Automatic partition pruning from metadata — but it only works if data is clustered. ORDER BY at load or clustering keys keep ranges tight.

STEP 3/6

BigQuery: Slots, Not Servers

BigQuery abstracts compute into slots — you think in query capacity, not machines.



On-demand: pay per byte scanned. Flat-rate/reservations: buy slot capacity for predictable workloads. Either way, zero cluster management.

- Columnar Capacitor storage + automatic sharding
- Dry-run queries to see bytes billed before running
- Partition + cluster tables to cut scanned bytes (and cost)

STEP 4/6

Zero-Copy Cloning

Instant, storage-free copies of tables, schemas, even whole databases.

A clone shares the underlying micro-partitions; only changed blocks consume new storage. CREATE TABLE … CLONE runs in seconds regardless of size.

- Dev/test environments on full production data, instantly
- Safe schema experiments: clone, alter, validate, swap
- Drops are cheap — clones inherit the retention window

STEP 5/6

Time Travel & Data Sharing

Query the past, share without copying.

Time Travel

AS OF timestamp / offset

Secure Data Sharing

no copies, no ETL

```
SELECT * FROM orders
  AT(OFFSET => -3600);
```

Share a live table with a partner account — they query, you govern.

Time travel (up to 90 days) makes 'oops' recoverable: undrop tables, audit history, reproduce yesterday's report exactly.

STEP 6/6

Cost Discipline

Elastic compute is a superpower and a bill shock. Guardrails matter.

- Auto-suspend warehouses after minutes of idle — always
- Separate warehouses per workload (BI vs ETL vs data science)
- Resource monitors: alert and auto-suspend at spend caps
- BigQuery: partition/cluster tables, avoid SELECT *
- Right-size: start small, scale up only when queries queue

The #1 cloud-warehouse failure mode is an XL warehouse left running over the weekend. Automation beats willpower.

STEP ✓/✓

Key Takeaways

Snowflake & BigQuery — Remember

- Storage and compute scale independently: cheap shared storage, elastic compute that suspends to zero.
- Micro-partitions with min/max metadata give automatic pruning — clustering keeps ranges tight.
- BigQuery's slots abstract away servers entirely; you manage query capacity, not clusters.
- Zero-copy clones create instant dev/test copies with no extra storage until data diverges.
- Time travel recovers mistakes; secure sharing distributes data without copies. Guard the spend with auto-suspend.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev